

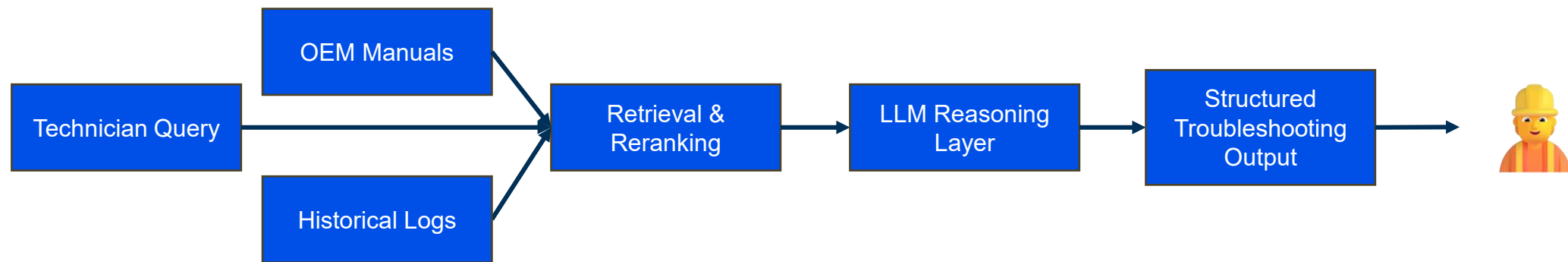


Traction Generative AI Internship Case

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Grounding Maintenance Decisions with OEM Manuals and Historical Context

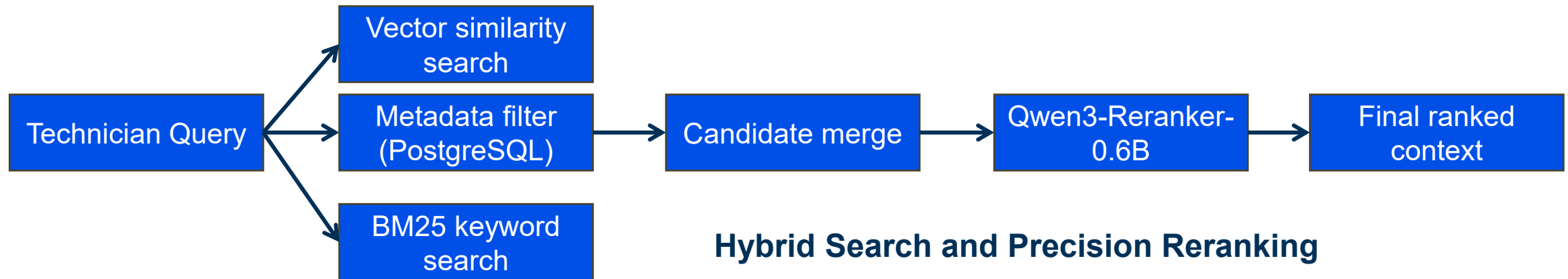
- A RAG system that combines two knowledge sources: (1) **static OEM manuals** and (2) **dynamic historical issue logs**; to give technicians instant, grounded troubleshooting guidance.
- At inference time, the system retrieves the **most relevant manual sections** and **similar historical incidents**, then produces **structured next-step guidance**.
- Supports 4 decisions: (1) inspection target, (2) applicable OEM procedure, (3) prior failure match, and (4) escalation/work-order need.
- Recommendations should appear as **contextual suggestions within the fault workflow**, with **proactive alerts** reserved for **high-confidence** or repeated failure patterns.



Retrieval-to-Decision Pipeline

RAG Ingest and Search Strategy

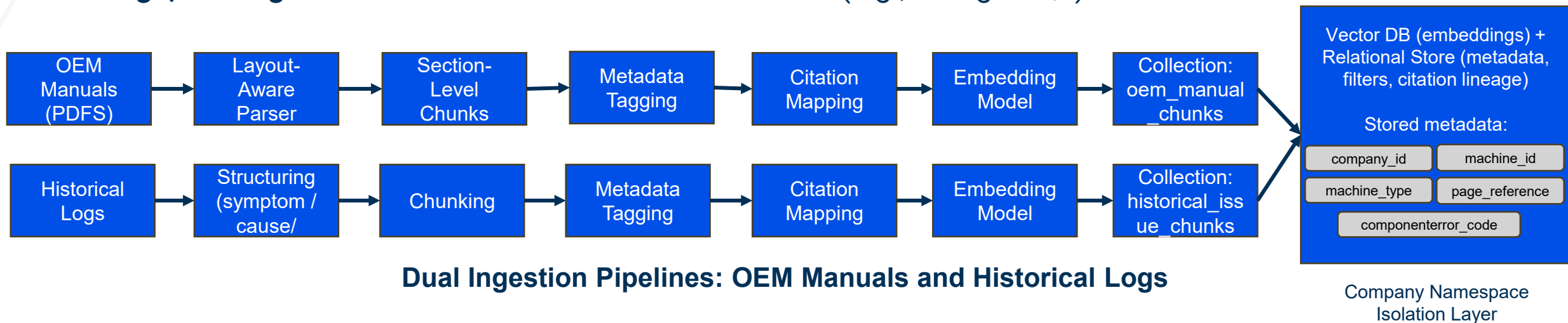
- OEM Manuals (PDF): Layout-aware parsing (e.g., Docling or Unstructured.io) that preserves section hierarchy, tables, and figures. Chunks are split at **section boundaries**, not arbitrary token counts. Images, charts, and diagrams are **linked as visual evidence alongside their parent text chunk**.
- Historical Logs (Markdown): Each issue is **normalized into structured fields**: machine ID, symptoms, root cause, action taken, outcome. Chunks are split at the **issue level** so each chunk captures a complete diagnostic episode.
- Embedding: **Qwen3-Embedding-8B** (MTEB score 70.58, Apache 2.0, 32K context, flexible dimensionality).
- Hybrid retrieval improves reliability over vector-only search, especially for technical queries containing error codes and part numbers (Xie et al., 2025; Alawadhi & Abbas, 2025).



Multi-Tenant Vector Database Architecture

- Two logical indexes serve different diagnostic roles:
 - oem_manual_chunks** — authoritative manufacturer procedures (static, versioned)
 - historical_issue_chunks** — plant-specific troubleshooting experience (dynamic, append-only)
- Separated indexes improve retrieval precision and update efficiency. **Modality-aware parsing** and **structured indexing strategies** are supported by manufacturing-domain RAG research (Li et al., 2026).
- Tenant isolation is enforced through **company-level namespaces (metadata partitioning)**. Retrieval never crosses customer boundaries. Queries are filtered hierarchically:

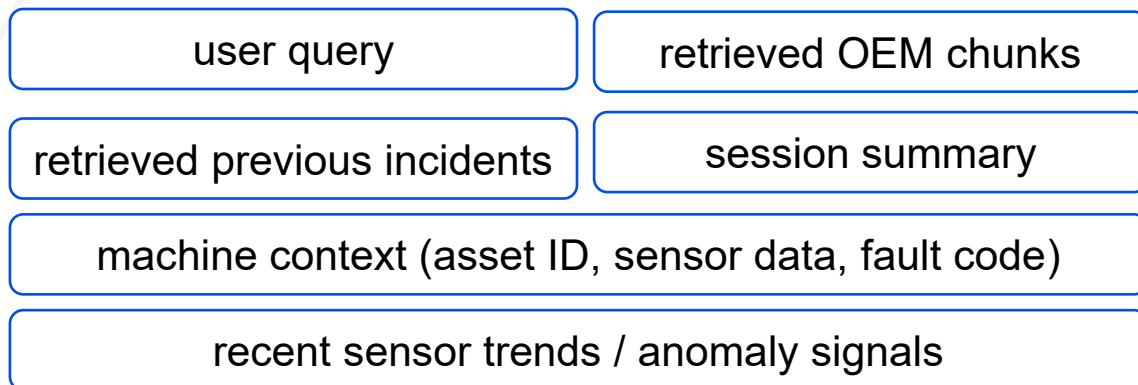
company_id → machine_type → machine_id → error_code
- Scalability: Embeddings** are stored in the **vector database** (e.g., Qdrant or Weaviate). **Raw text, citation lineage, and ingestion metadata** live in a **relational store** (e.g., PostgreSQL)



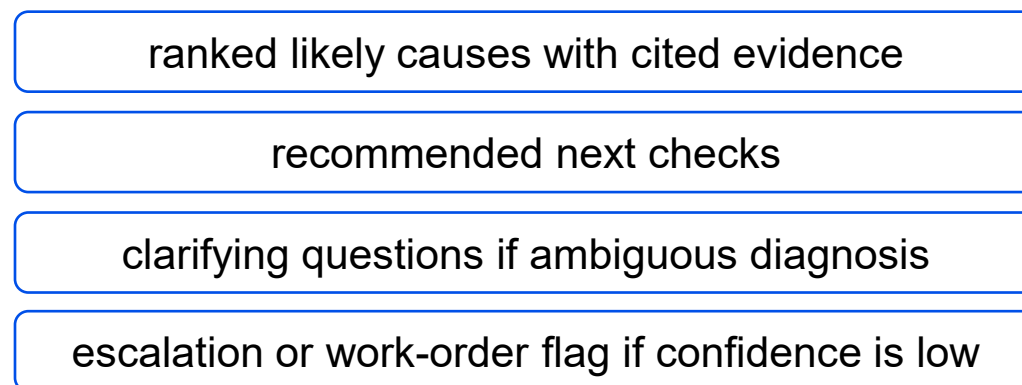
Agent Workflow, Session Storage, and Structured Outputs

- **Agent Architecture:** A graph-based workflow (LangGraph) with deterministic nodes, not a free-form chatbot. Troubleshooting requires controlled retrieval, grounded synthesis, and repeatable step-by-step reasoning.
- Session Storage — two memory layers:
 - Short-term: the **most recent N turns** of the troubleshooting conversation
 - Rolling summary: a **compressed record of what has been checked**, ruled out, and recommended so far
- This keeps context **within token limits** while preserving critical diagnostic state.

Inputs at inference



Structured outputs



Graph-Based Agent Workflow with Session Memory



Integration, Risks, and Why This Matters for Traction

- The assistant is embedded in existing surfaces: the **machine page**, the **fault view**, and the **work-order interface** (not a standalone chatbot)
- Guidance surfaces at three moments: when a **sensor alert fires**, when a technician **opens a machine fault**, and when the system detects a **similar historical incident**. Recommendations include **relevant manual sections**, **similar past resolutions**, and **suggested next steps**.

Risks	Mitigations
Hallucinated repair steps	Retrieval-grounded responses only, with explicit source citations
Incomplete OEM documentation	Confidence thresholds ; recommendations only surface when evidence is sufficient
Retrieval mismatch across machine variants	Hierarchical metadata filtering by machine type and model
Noisy historical logs	Structured normalization at ingest; technician confirmation before work-order updates

- These risks are acceptable as recommendations remain **assistive**, **traceable**, and **human-validated**.
- This RAG tool converts a sensor alert into a ranked list of likely causes backed by OEM procedures and historical precedent so the technician acts on structured evidence instead of searching through PDFs and unstructured logs.

